

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3445U10-1



S18-3445U10-1

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment

FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	13	
3.	10	
4.	13	
5.	9	
6.	10	
7.	9	
Total	75	

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01

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3(b)(i)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 24.

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Answer **all** questions.

Examiner
only

1. Raw materials are extracted from the Earth's crust or atmosphere by different methods.

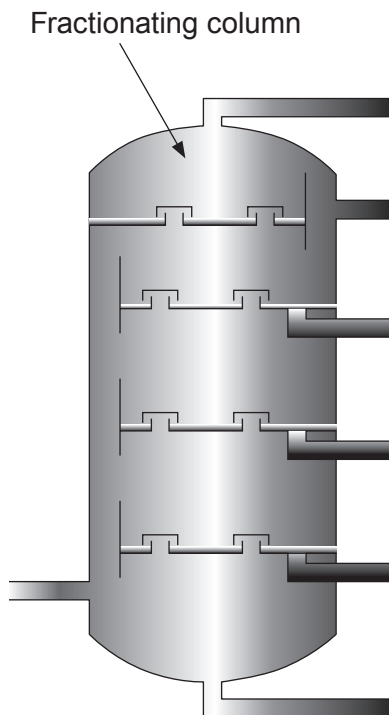
- (a) Identify the correct extraction method for each raw material by drawing a line.
One has been done for you.

[2]

Raw material	Extraction method
crude oil	fractional distillation
shale gas	drilling
haematite (a metal ore)	surface mining
oxygen	fracking

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03

- (b) A fractionating column is used to separate the hydrocarbons in crude oil. The table shows information about some of the fractions obtained.



Fraction	Carbon chain length	Boiling range (°C)
Refinery gases	1-4	<40
Petrol	5-8	40-110
Naphtha	8-10	110-180
Kerosene	10-16	
Diesel	16-20	260-320
Fuel oil	20-50	320-400
Bitumen	>50	>400

(i) **Complete** the table. [1]

(ii) Name the process which is used to separate crude oil into its fractions. [1]

.....

(iii) State what happens to the crude oil immediately before it enters the column. [1]

.....

(iv) State the fraction that a hydrocarbon with a boiling point of 290 °C will belong to. [1]

.....

(c) Complete the statements below by underlining the correct term in the brackets.

(i) The temperature (**increases, stays the same, decreases**) going up the fractionating column. [1]

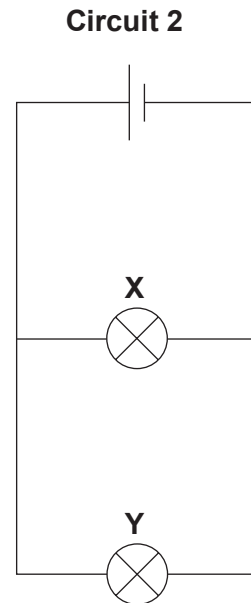
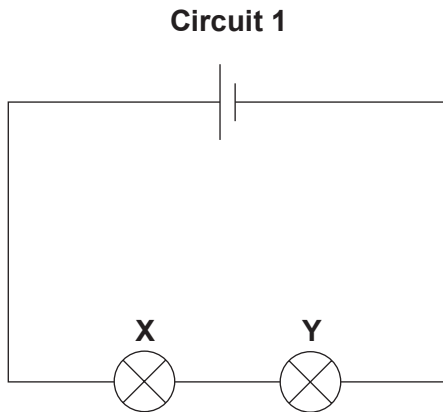
(ii) The boiling point of a hydrocarbon (**increases, decreases, stays the same**) as the chain length increases. [1]

(d) The fractionating column produces hydrocarbons that can be further processed and used to make plastics.

Complete the table to show how polyethene, polychloroethene (PVC) and polytetrafluoroethene (PTFE) are made. [3]

Monomer			Polymer	
Name	Molecular formula	Structural formula	Name	Structure
Ethene	C_2H_4	$ \begin{array}{c} H & & H \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ H & & H \end{array} $	polyethene	
Chloroethene	C_2H_3Cl		PVC	$ \left[\begin{array}{cc} Cl & H \\ & \\ -C & -C- \\ & \\ H & H \end{array} \right]_n $
Tetrafluoroethene		$ \begin{array}{c} F & & F \\ & \diagdown & / \\ & C = C \\ & / & \diagdown \\ F & & F \end{array} $	PTFE	$ \left[\begin{array}{c} F \\ \\ -C- \\ \\ F \end{array} \right]_n $

2. A student is studying the properties of different electrical circuits. **X** and **Y** are identical lamps. The student built the following circuits.



- (a) Complete the sentences using the terms in the box.
The terms can be used once, more than once or not at all.

series	less than	parallel	the same as
greater than	simple	current	voltage

- (i) The lamps in **circuit 1** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]
- (ii) The lamps in **circuit 2** are connected in
 The current through lamp **X** isthe current through lamp **Y**.
 [2]

(b) In **circuit 1** the voltage supplied is 9 V. Each lamp has a resistance of 3 Ω.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in this circuit.

[3]

Current = A

(ii) State what would happen to the current in **circuit 1** if lamp **Y** is removed and the circuit was built with only lamp **X**. [1]

.....

(c) In **circuit 2** the voltage supplied is also 9 V and each bulb has a current of 3 A.

Use the equation

$$\text{power} = \text{voltage} \times \text{current}$$

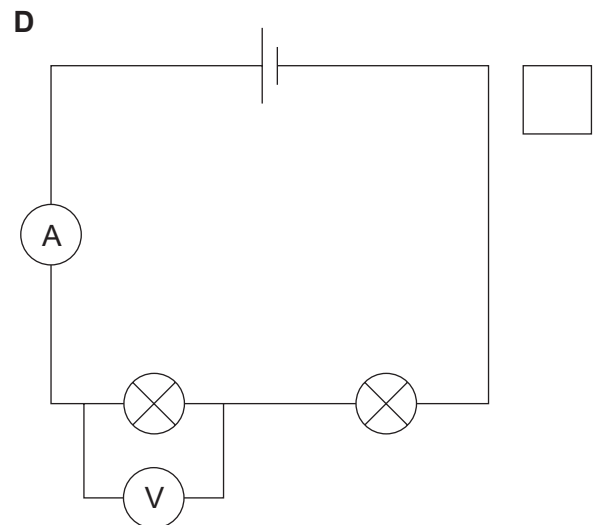
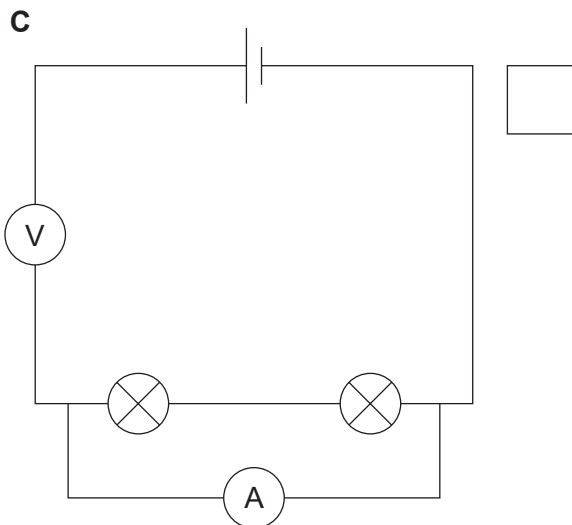
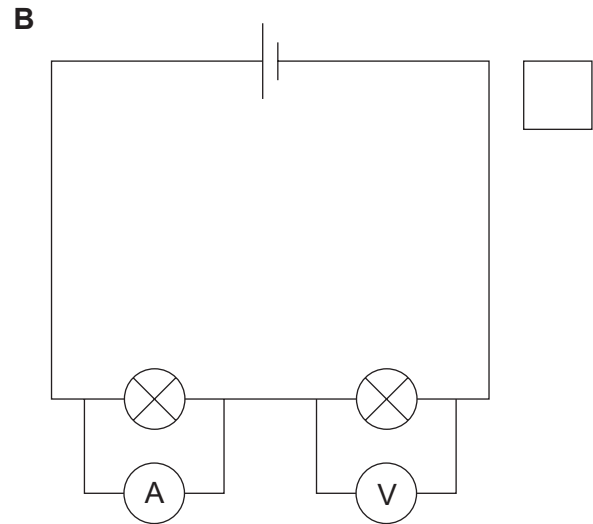
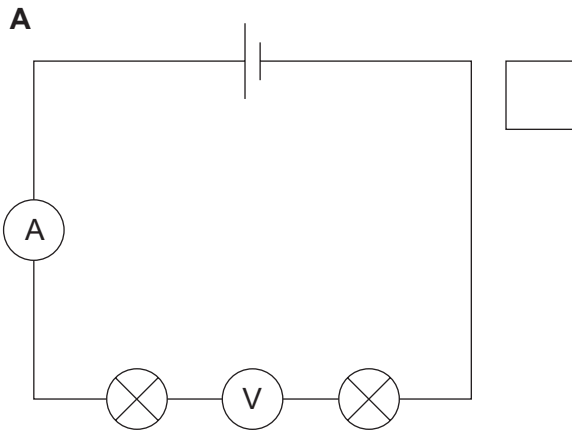
to calculate the power of lamp **X**.

[2]

Power = W

- (d) The student set up a circuit to measure the current and voltage correctly.

Use a tick (✓) to identify the circuit diagram that should have been used to measure the voltage. [1]



- (e) State **two** reasons why lamps are connected in parallel rather than in series in the home. [2]

1.
2.

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3. Water treatment in the UK is an important process in making water safe to drink.

- (a) In an experiment to determine the hardness of water in different parts of the country, different technicians tested water samples from Bristol, Birmingham, Manchester and Southampton.

They divided the water samples from each city into two and carried out Test A and Test B on each portion.

Test A

For the **first** portion they followed the method below.

1. Measure 15 cm^3 of water sample into a boiling tube.
2. Add 1 cm^3 of soap solution, insert a stopper then shake vigorously for 5 seconds.
3. Repeat step 2 until lather forms that lasts for 30 seconds. Record the total volume of soap solution added.
4. Repeat steps 1-3 with 15 cm^3 of the other samples of water.

Test B

For the **second** portion they followed the method below.

- Boil the water for 10 minutes.
- Then follow steps 1-4 in **Test A**.

- (i) State the dependent variable in the experiment.

[1]

.....

- (ii) How could the reproducibility of the experiment be checked?

[1]

.....

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- (b) The results of the experiment are shown below.

Sample	Volume of soap used to form lather (cm ³)	
	Test A (without boiling)	Test B (after boiling)
Birmingham	1	1
Southampton	11	1
Bristol	12	12
Manchester	1	1

- (i) Explain how the technicians used the results to make conclusions about the type of water in **each city**. [6 QER]

- (ii) Hard water can cause problems in the home.

State **two** problems hard water can cause.

[2]

1.
2.

4. Enzymes are involved in all metabolic reactions.

(a) (i) State the **type** of biological molecule that all enzymes are made of.

[1]

.....

Enzymes can break down larger molecules into smaller units in the digestive system.

mouth	amylase	lipase	liver
large intestine	amino acids	fibre	stomach

Enzyme	Larger molecules	Smaller units	Site of action
Carbohydrase	Starch	Glucose	 and small intestine
.....	Fats and oils	Fatty acids and glycerol	small intestine
Protease	Protein		 and small intestine

(ii) **Complete the table** using only the words in the box.

[4]

- (b) Students investigated the effect of temperature on the digestion of starch.

The students mixed starch and amylase (a carbohydrase) at different temperatures.

They used iodine to test for the presence of starch. Iodine changes colour from brown to blue-black if starch is present.

The results of the students' investigation are shown below.

Temperature of the solution (°C)	Colour of the solution					
	at start	after 1 min	after 2 min	after 3 min	after 4 min	after 5 min
20	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black
30	blue-black	blue-black	blue-black	blue-black	blue-black	brown
40	blue-black	brown	brown	brown	brown	brown
50	blue-black	blue-black	brown	brown	brown	brown
60	blue-black	blue-black	blue-black	brown	brown	brown
70	blue-black	blue-black	blue-black	blue-black	brown	brown
80	blue-black	blue-black	blue-black	blue-black	blue-black	blue-black

- (i) Describe how temperature affects the digestion of starch between **20 °C** and **70 °C**. [3]

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- (ii) Explain the results at **80 °C**. [2]

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(c) When starch is completely broken down by amylase the final product is glucose.

(i) Describe the test that the students need to perform to show the presence of glucose. [2]

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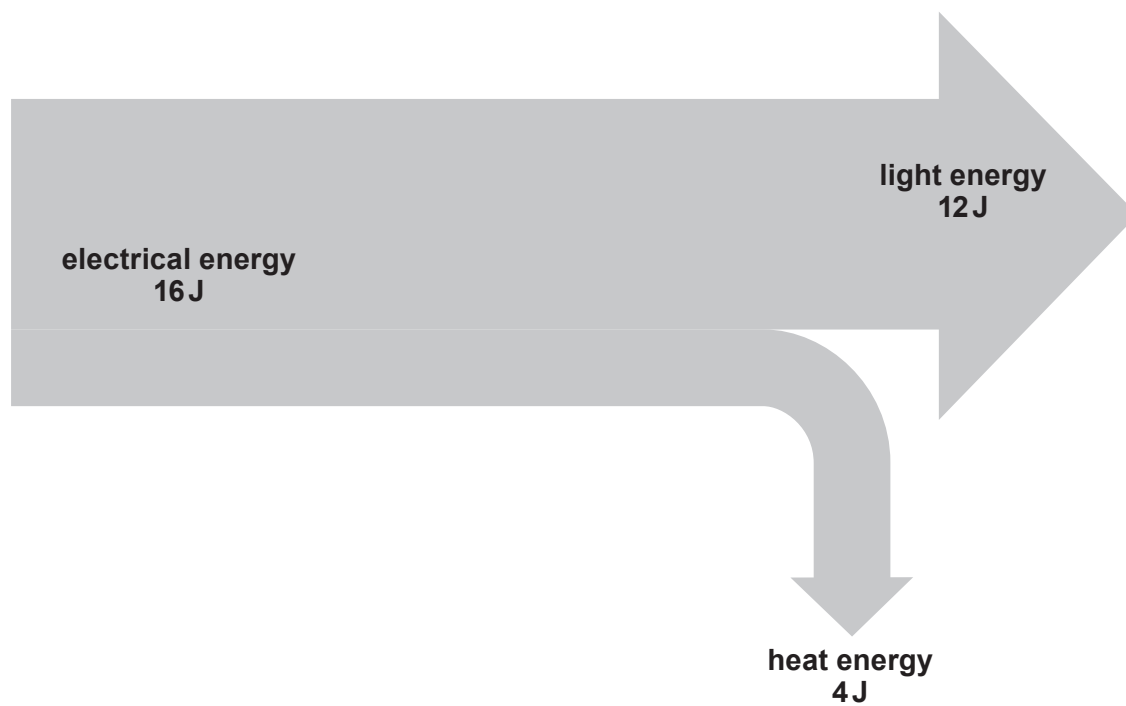
(ii) State the colour **change** that shows that glucose is present. [1]

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5. Lighting a home accounts for 18 % of a typical house's electricity bill.

(a) The diagram below shows energy flow through a LED lamp.



Use the equation:

$$\% \text{ Efficiency} = \frac{\text{energy usefully transferred}}{\text{total energy}} \times 100$$

to calculate the efficiency of the LED lamp.

[2]

% Efficiency =

- (b) The Energy Trust is an organisation in the UK that encourages householders to become more efficient in their energy use and to help to reduce their energy costs. The table below compares traditional filament and LED lamps.

	LED	Traditional filament lamp
cost of 1 lamp	£2.50	£1
lamp lifespan (hours)	50 000	1000
number of lamps used in 50 000 hours	1	
cost of lamps used in 50 000 hours	£2.50	£
power per lamp (kW)	0.01	0.06
units used in 50 000 hours (units = power $\times$ 50 000)		3 000
cost of one unit	20p	20p
cost of electricity for light for 50 000 hours = units used $\times$ cost of unit	£	£600
total cost of light for 50 000 hours	£102.50	£

- (i) **Complete** the table.
(space for working)

[5]

- (ii) The homeowner refuses to buy LED lamps because they are more expensive and he thinks he will lose money compared to using traditional filament lamps.

Explain whether you agree with the homeowner.

[2]

.....

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6. Students are making the fertiliser potassium nitrate in the laboratory. Most fertilisers are formed by the reaction of an acid and an alkali.

(a) Complete the word equation for the formation of potassium nitrate. [2]

..... + potassium hydroxide $\longrightarrow$ potassium nitrate +

- (b) To make the fertiliser, it is important that the students are able to exactly neutralise the acid with alkali. To do this the students carefully measured 25 cm^3 of acid into a flask and titrated just enough alkali to neutralise the acid.

The results of the experiment are shown below.

Trial	Volume of alkali added at endpoint (titre) (cm^3)
1	20.5
2	20.8
3	18.5
4	20.0
5	19.9

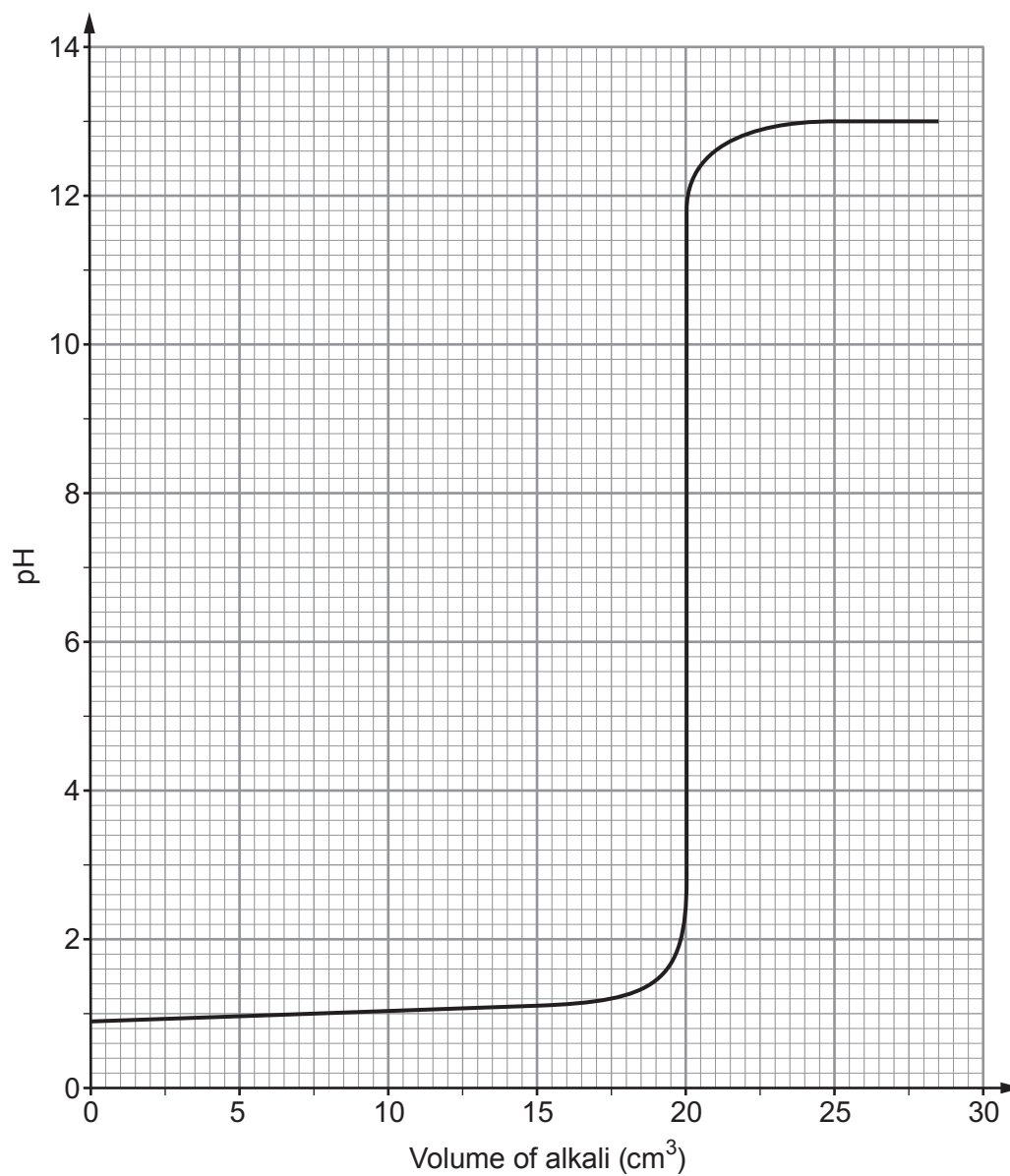
- (i) The reading for trial 3 was considered to be an anomalous result. Give a reason why. [1]

.....

- (ii) Calculate the mean titre. [1]

Mean titre = cm^3

- (c) The graph below shows the pH changes, measured by a pH probe, that occurred during trial 4 of the experiment.



Explain the pH changes shown on the graph.

[3]

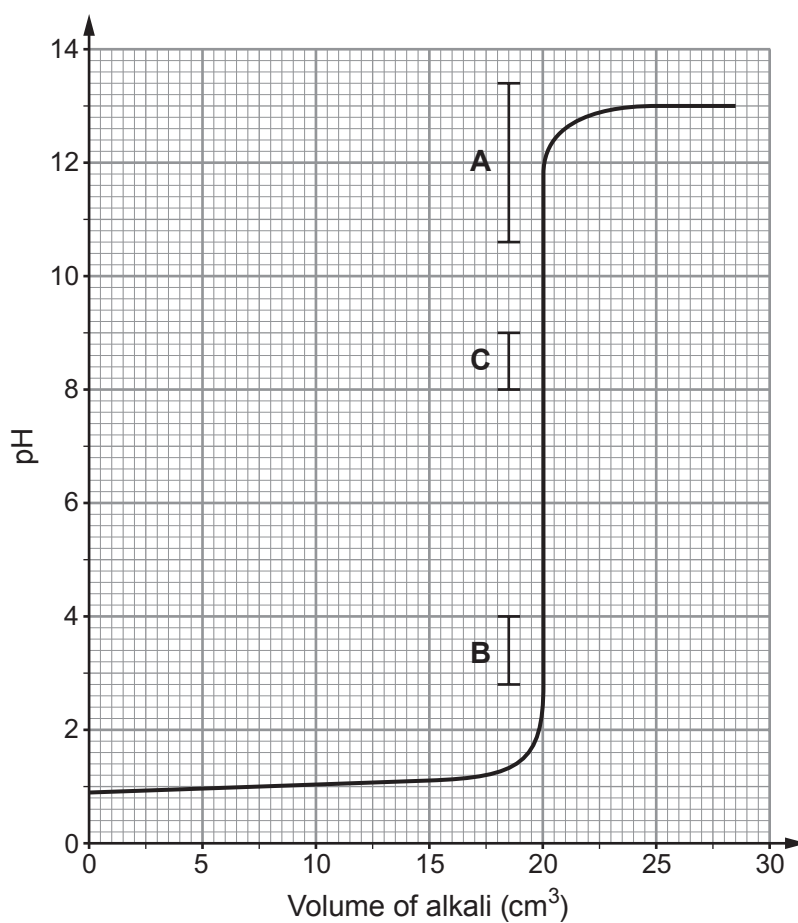
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- (d) Not all the students had a pH probe. Some had to use an indicator. The pH range of three different indicators is marked on the graph as **A**, **B**, and **C**.



- (i) Use the information in the table below to match the range to each indicator. [1]

Indicator	pH range	Colour change	Range on the graph (A , B , or C ?)
methyl blue	10.6 - 13.4	blue to violet	
phenolphthalein	8.0 - 9.0	colourless to pink	
dinitrophenol	2.8 - 4.0	colourless to yellow	

- (ii) Explain which indicator is not suitable for this experiment. [2]

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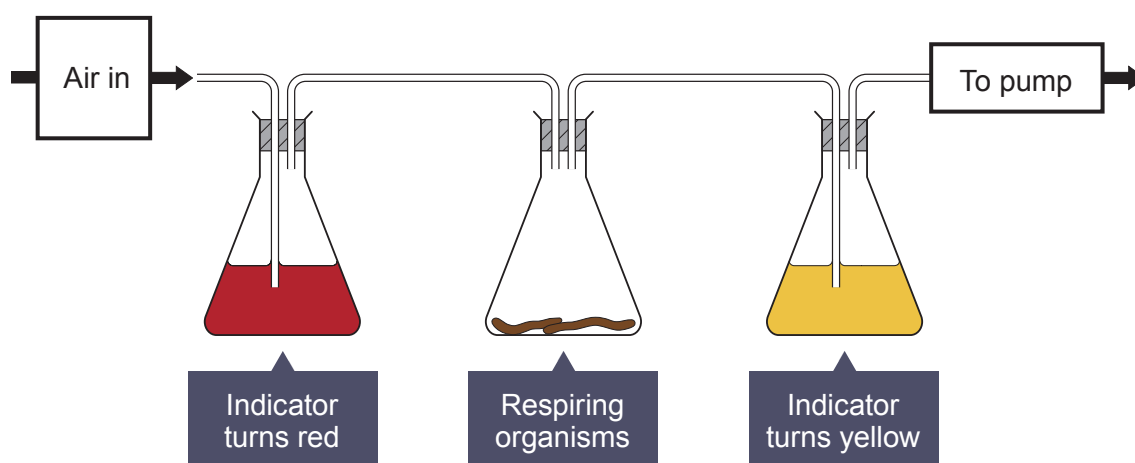
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7. Hydrogen carbonate indicator can be used to measure the rate of respiration.

Concentration of carbon dioxide (%)	Colour of hydrogen carbonate indicator
less than 0.04	purple
0.04 (normal concentration in air)	red
greater than 0.04	yellow

The apparatus below can be used to compare the rate of respiration in small animals.



A group of students used different respiring organisms and recorded the time taken for the hydrogen carbonate indicator to turn yellow. The apparatus was kept at a constant temperature of 20 °C.

The results are shown in the table below.

Organism	Time taken for the hydrogen carbonate indicator to turn yellow (min)
earthworm	22
woodlouse	18
grasshopper	10
mouse	3

- (a) Explain why the hydrogen carbonate indicator changed colour in this experiment. [2]

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.....

- (b) Give **two** reasons why the hydrogen carbonate indicator turned yellow fastest with the mouse. [2]

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- (c) The experiment was performed at a temperature of 20 °C. [2]

State **two** other variables that need to be controlled when comparing the respiration rates of different organisms.

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- (d) Describe how you would set up a control for this experiment and state the expected result. [3]

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END OF PAPER

THE PERIODIC TABLE

Group

2
1

07

1 H Hydrogen 1																																			
7	Li	9	Be							11	B	12	C	14	N	16	O	19	F	20	Ne														
	Lithium		Beryllium								Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon														
23	Na	24	Mg							27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar														
	Sodium		Magnesium								Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon														
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	106	Rh	103	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Rhodium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	195	Ir	192	Pt	197	Au	201	Hg	204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Thallium		Lead		Bismuth		Polonium		Astatine		Radon
223	Fr	226	Ra	227	Ac																														
	Francium		Radium		Actinium																														

Key

Key

relative atomic mass

Symbol	Name	atomic number
A_r		